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Original Research

Principles and attributes of evidence-based co-creation: From naïve praxis toward a trustworthy methodology - A Health CASCADE study

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ABSTRACT

Objectives: Co-creation is recognised as a promising methodology for addressing complex public health issues by leveraging the collective intelligence of multiple stakeholders to develop tailored solutions. However, there is a lack of clarity and significant fragmentation in the concept. This study aims to define the fundamental attributes and guiding principles of co-creation to establish it as a rigorous, evidence-based, and trustworthy methodology.
Study design: A participatory concept mapping process based on the Differentiae Principle was employed.
Methods: The study consisted of four stages: 1) systematic extraction of attributes and their definition from traditional scientific and participatory research methodological literature, 2) screening, 3) semantic analysis and clustering into principles using Natural Language Processing and Large Language Models and 4) selection of a final set of attributes and development of principles. The study involved 50 participants with diverse backgrounds within and outside the Health CASCADE Network over 18 months.
Results: The study identified 105 candidate attributes. A set of 38 was selected from which 10 key principles were derived (open access, recognised contribution, ethics and legal, transparency, rigorous evaluation, clear problem, plural evidence, critical reflection, diversity, and structured and flexible processes) across three domains: Governance, Scientific and Methodological Rigour, and Collective Intelligence Processes.

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Conclusions: The attributes and guiding principles provide a comprehensive foundations for evidence-based co-creation, enhancing its trustworthiness, transparency, reliability, and impact in public health and other fields. Future research and practice should use these principles and attributes to refine and implement co-creation effectively.

1. Introduction

The increasing complexity of societal challenges—including public health and planetary health crises—demands innovative and inclusive knowledge generation and application approaches.[1] In recent years, co-creation has gained traction as a promising methodology to bridge the gap between research and practice to tackle complex and so called 'wicked' problems[2] (such as obesity) requiring transdisciplinary collaboration in public health and many other fields.[3] By engaging multiple stakeholders in the research process, co-creation aims to foster contextually relevant, socially acceptable, and actionable knowledge. It can potentially enhance both the effectiveness and impact of solutions developed, particularly for complex public health issues, which often pose challenges in resolving, especially for specific populations.[4]

Despite its growing popularity, co-creation remains conceptually and methodologically fragmented, lacking a unified methodology that ensures both scientific rigour and real-world impact.[5] Critics argue that co-creation is not a coherent methodology but has become an umbrella term encompassing diverse methods without coherent methodological foundation[5] The exponential proliferation of published definitions over the past decade has paradoxically increased rather than resolved conceptual ambiguity.[6,7] This proliferation stems from three key sources of confusion: first, researchers from different disciplines have developed field-specific definitions without cross-disciplinary dialogue; second, many definitions conflate co-creation with related participatory research approaches; and third, most definitions focus on aspirational outcomes rather than operational characteristics that can be implemented and evaluated.[8] Although semantically similar, these definitions remain frustratingly vague, offering broad perspectives on collaboration and democratic participation while failing to specify actionable attributes. The resulting discourse has prioritised normative and moral debates about authentic co-creation over systematic efforts to establish core principles and methodological standards.[6,7,9] Consequently, co-creation has not evolved into a scientifically defined, structured, and rigorous methodology that researchers and practitioners can systematically apply across contexts. While a definition is important, a single sentence is insufficient to fully characterise and delineate a concept as multifaceted as co-creation[10] or to resolve its ambiguities. Instead, establishing a strong methodological and conceptual foundation requires systematically outlining the desirable attributes of co-creation—that must be translated, acted upon, and upheld in practice. We argue that co-creation must evolve into an evidence-based methodology that integrates the two dominant modes knowledge production:[11,12] traditional, disciplinary scientific research (Mode 1); and, a more collaborative, context-sensitive, and problem-oriented approach (Mode 2). To achieve this, we move beyond definition alone to explicitly delineate the fundamental intrinsic attributes and guiding principles necessary to establish co-creation as a rigorous, trustworthy, participatory and evidence-based scientific practice for bridging the gap between knowledge and action in public health. This study aims to establish the methodological attributes and principles for co-creation by a) define the fundamental attributes of co-creation that fully delineate it and b) derive principles that govern co-creation.

2. Methods

2.1. Study Design

We followed a participatory concept mapping[13] process based on the Differentiae Principle.[14] The study followed a pre-defined protocol comprising of 5 stages: 1) the systematic extraction of attributes (differentiae) from Mode 1 and Mode 2 research, 2) informed screening of attributes using the Differentiae Principle, 3) semantic analysis of differentiae and clustering of characteristics into principles and domains, 4) selecting a minimal set of attributes, and 5) definition of principles.

This methodological decision was deliberate and reflects our commitment to establishing co-creation as a rigorous scientific methodology. It combines: 1) systematic extraction of foundational scientific and research attributes from methodological literature to ensure comprehensive theoretical grounding, 2) collective intelligence and wisdom of experience from diverse participants, and 3) the rigour of set theory and the Differentiae Principle to maintain conceptual precision and avoid the methodological dilution and conceptual dilution and ambiguity prevalent in the literature.[8]

We deliberately work across domains and fields to ensure that the principles and attributes derived would be broadly applicable and foundational.

The process is detailed below and was conducted collectively throughout 18 months, starting in September 2021 and culminating in a plenary meeting of the Health CASCADE Network in June 2023. At each stage, a report was circulated through the network of participants for transparency, both keeping participants informed and for member checking.

2.2. Participants

The members of the Health CASCADE network conducted this piece of collective work. Health CASCADE is a scientific network bringing together pluridisciplinary and multisectoral experiences of working with co-creation and participatory research methods. Health CASCADE mission is to establish co-creation as a robust scientific methodology for developing solutions to complex problems in public health and other fields.[11] The Health CASCADE network consists of 50 academics from seven European Institutions, and 12 affiliated academics from outside the EU at different stages of their careers (amongst those there were: 15 Marie Skłodowska-Curie Doctoral Fellows, 15 Full Professors, 8 Early Career researchers) from varying disciplines including Anthropology, Architecture, Behavioural Sciences, Cognitive Sciences, Computer Sciences, Social Sciences, Social Policy, Gerontology, Health Promotion, Human Geography, Movement and Sport Sciences, Public Health, Physics, Physiotherapy, Psychology, Urban design. In addition, the participants also included 30 stakeholders in co-creation from 28 organisations in different sectors, from public health agencies to community groups. Overall, the participants were 80 % female, between 19 and 62 years old, and from diverse nationalities across the six inhabited continents. Further details of the network can be found on www.healthcascade.eu. All participants were considered and treated as co-researchers[15] in this process.

Stage-specific participation was as follows.

- **Stage 1 (Literature extraction):** 15 academic co-researchers conducted the systematic extraction from university curricula, as this required access to institutional reading lists
- **Stage 2 (Screening):** 31 co-researchers (both academic and non-academic) participated in the Amsterdam workshop, with 35 participating in online member-checking

- **Stage 3 (Semantic analysis):** Conducted by the full network through circulation and feedback
- **Stage 4 (Selection):** 28 co-researchers from across the network provided individual parsimonious sets
- **Stage 5 (Finalisation):** The full network participated in the Uppsala meeting and final consensus process

2.3. Differentiae Principle

The differentiae principle stems from Aristotelian logic. It is a process for defining a concept by identifying a set of specific attributes that collectively differentiate the concept from others.[14] It is a fundamental methodology for clarifying the essential attributes that make a concept unique, aiding in clear communication and understanding.[14] The differentiae are the elementary units from which the meaning of a concept is built. They represent fundamental attributes of the concept. A set of differentiae defines a broader genus (category), which, here, is a guiding principle. The genera (plural) and underlying differentiae then fully delineates the concept. This means that a differentia cannot be removed from the definition of a concept without changing its nature.[14]

2.4. Systematic extraction of the differentia

Fifteen researchers from the Health CASCADE network systematically extracted attributes fundamental to and defining of Mode 1 and Mode 2 research from methodological texts. The text were identified by the network from the reading list of the Universities involved in the network (Ghent University, Glasgow Caledonian University, Blanquerna-Universitat Ramon Llull, Umea University, Amsterdam UMC, Constructor University) that they revommended as foundational texts in their curriculum on research methods in Mode 1, implementation science, translational research, and mode 2 e.g. participatory research texts.[16–25] This literature-based extraction was deliberately chosen to ensure comprehensive coverage of research and scientific methodological principles from both traditional scientific research and participatory approaches, providing more grounded foundation than experiential knowledge alone could. The fifteen researchers were trained and instructed to semantically identify attributes as adjectives (lexical words describing and providing information about the qualities and characteristics of something) such as “unbiased”, “reproducible”, or “democratic”. Researchers also extracted the definition or explanation of the attributes, as well as the methods or ways by which they are enacted in practice. For example, the attribute “unbiased” is enacted through “random sampling”, “using randomised control trials”, and “peer review process”, amongst others.

Then, in four groups of three to four researchers each, the team discussed the extracted attributes and combined them into a list, alongside the extracted definition(s) and method(s) of enactment. The list was distributed through the whole Health CASCADE Network for member checking and completeness.

2.5. Screening

The screening of attributes extracted, after deduplication, took place during an in-person workshop in Amsterdam in September 2022 ($N = 31$) and was followed by an online member-checking ($N = 35$) event to ensure that the final list represented the participatory decision-making process. At this stage, the experienced co-creation researchers applied their collective expertise to evaluate the literature-derived attributes, determining relevance, adding missing attributes and resolving ambiguities through their practical knowledge. The decision–making process used an informed consent process;[26] decision was taken unless there was an objection. The attributes were classed initially by the group as: (a) completely irrelevant, redundant or uninformative, (b) universal (attributes universal to all modes of research and therefore relevant to co-creation), and (c) maybe (attributes that could be defining co-creation). Disagreements were settled through debate until there were no objections to the classification of an attribute in these three categories.

2.6. Selection and deriving principles

Participants were asked to assemble, independently, individually and in their own time, the most parsimonious set of attributes they would want co-creation to have from the selected pool of attributes. This approach combined the rigor of systematic literature extraction from foundational texts with collective intelligence and wisdom of practical experience, guided by a systematic differentia method, thus ensuring the sets of attributes was both theoretically complete and practically validated. Participants were asked to summarise the attribute, before writing justifications for the inclusion or exclusion of the attribute based on the differentiae. They were given six weeks to allow for reflection. Twenty-eight researchers from the network provided full answers and parsimonious sets.

We analysed the responses using Natural Language Processing (NPL)[27] to.

a) Gauge the semantic similarity of responses between the respondents and identify attributes that appeared controversial or had divergent reasons.
b) Form clusters of attributes under a set of general principles using the Jaccard semantic dissimilarity index approach[28]
c) Draft principles based on topic modelling of the attributes definition and participants’ responses using Latent Dirichlet Allocation[29] and then cluster the principles (genera) into domains.

Two researchers (SC and NS) then used Large Language Models (LLMs), specifically ChatGPT-3.5, to repeat this analysis for cross-validation as well as generate summary definitions for all attributes and multiple versions of the principles definitions (genera) and overarching themes (domains).

2.7. Finalising the attributes, principles and domains

The analysis above was shared with the whole Health CASCADE network. In June 2023, the network met in person to review the analysis and finalise the attributes and principles of evidence-based co-creation. First, the network was invited to comment on the results of the analysis, specifically discuss attributes that were controversial, those that had weak reasons for exclusion or inclusion, and those that did not cluster well within a genus. Second, members of the network in groups of four created their own set of attributes (from the pool of selected attributes) and sorted these under a set of principles/genera (Fig. 1). Each group conducted a stress test by sequentially removing each attribute from their set one at the time, questioning whether this set was sufficient to delineate evidence-based co-creation, and questioning whether any attributes were redundant. The procedure was repeated until they could no longer remove or move attributes around the genera. Finally, we stress-tested the sets using an inversion reasoning method.[30] We asked network members to write down “what they did not want evidence-based co-creation to be”. They then used these statements to check that the set of attributes and principles they selected was coherent against this counter definition. Results were synthesised, and the final set of attributes and principles, together with their definitions, agreed upon using the informed consent method.[29]

3. Results

The information flow is depicted in Fig. 2. We identified 105 attributes, amongst which there were 13 duplicates. After screening, we included 73 unique attributes. Next, six attributes were excluded during the reasoned selection process, ten were excluded through semantic

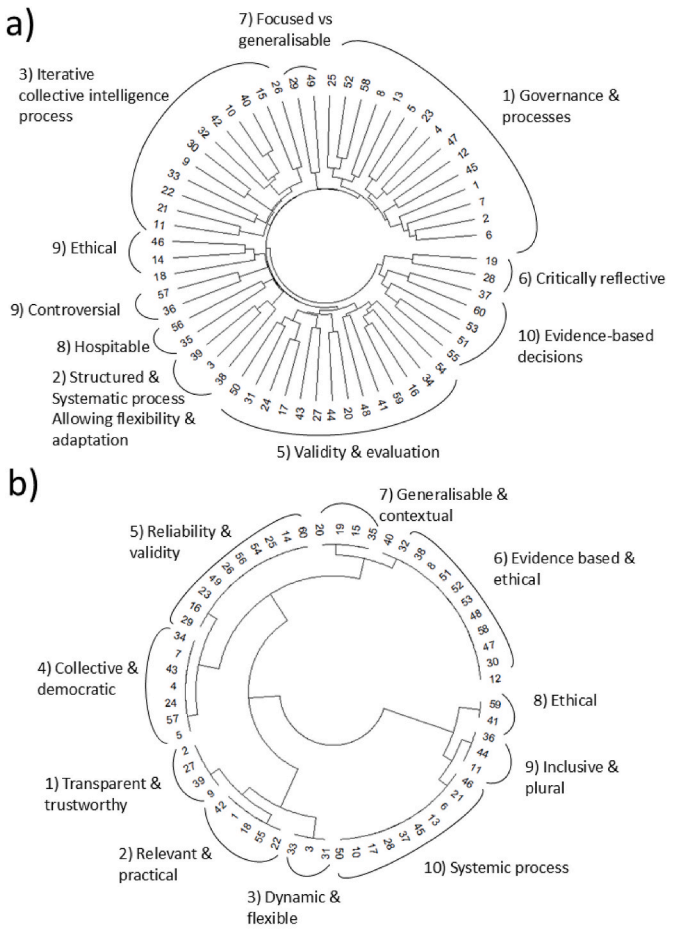


Fig. 1. a) Clustering of the reason participants gave for the inclusion of attributes derived from natural language processing and semantic analysis (numbers refer to attributes see supplementary 2 table S2.3) b) Clustering of the definition of each attribute derived from natural language processing and semantic analysis (numbers refer to attributes see supplementary 2 table S2.2) and c) Uppsala network meeting to finalise the set of attributes and principles.

analysis, and nineteen were deemed redundant during the creation of the parsimonious set and inversion reasoning check. Fig. 3 represents a robust taxonomic structure where 38 attributes form 10 general principles, which in turn constitute three domains, derived through semantic analysis, set theory and collective human intelligence.

The final set of attributes characterising evidence-based co-creation and the principles derived from them is presented in Fig. 3. The definition of each of these attributes is available in supplementary material S1. The full set of identified attributes is available in the supplementary materials S2.

The principles that were derived from the definition of the attributes fall into three domains: 1) Governance, 2) Methodological and Scientific rigour, and 3) Collective intelligence.

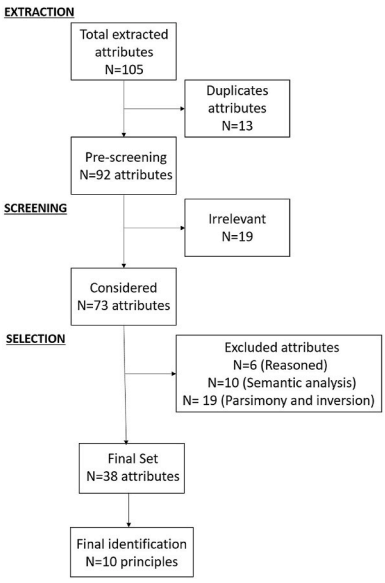


Fig. 2. Information flow.

3.1. Governance

1) **Open access:** Co-creation should prioritise open access to knowledge and information, including practical protocols, data, publications, and other outputs, wherever possible. This can help ensure that the process is transparent and accessible to all stakeholders.
2) **Recognised contribution:** Participants in co-creation should be acknowledged and recognised for their contributions, whether through co-authorship on publications, intellectual property, attribution in reports, or other means. This can help build trust and credibility with participants and stakeholders.
3) **Legal and ethical considerations:** Co-creation should govern legal and ethical considerations related to copyright, intellectual property, data sharing agreements, confidentiality, attribution, and the environmental impact of any activities. Co-creation should adhere to all principles of research ethics by integrating ethical reflexivity in the co-creation process within a hospitable environment. This can help ensure that the process is conducted in a responsible and ethical manner.
4) **Transparency:** Co-creation should be transparent and open throughout the process, from problem definition to solution development and implementation. Stakeholders and co-creators should be fully informed in clear and easy language of the rules of engagement and the process. This can help build trust and credibility with participants and stakeholders.

3.2. Scientific and methodological rigour

5) **Clear problem:** Co-creation should focus on a clear problem or challenge, taking into account the boundaries with respect to resources and time. The problem must be pragmatic and relevant to the community of stakeholders involved.
6) **Plural evidence:** The co-creation process must be based on the best available evidence from the widest possible plurality of sources, including scientific research, local knowledge, and other relevant information. In addition, the process must use evidence-based co-creation methods where possible.
7) **Reflection:** Co-creation is iterative, and critical reflection must be integrated into every step of the process, based on continuous feedback and evaluation, to ensure that it remains relevant, meaningful, ethical and sustainable.



Fig. 3. Attributes of co-creation (Outer ring), the Principles derived from them (Middle ring) and Domains (Inner ring and colour code). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

8) **Rigorous evaluation:** The co-creation process must be protocol-based including a continuous process evaluation. In addition, the co-created solutions should be evaluated for their contribution to solving the problem, as well as their impact on participants and the broader community of stakeholders (e.g. ripple effects). This evaluation should be based on robust and rigorous methods that are transparent and open to scrutiny.

3.3. Collective intelligence

9) **Structured and flexible process:** Co-creation must be based on a structured collective intelligence process that is inclusive, interactive, and socially coherent, dynamic and misadaptable. Therefore, it must be planned and conducted systematically with flexibility in mind: *"Plan the plan and plan how to change the plan!"*.
10) **Diversity:** Co-creation must engage the widest possible diversity of stakeholders, including those who are traditionally represented, and ensure that all voices/perspectives/knowledge are heard and valued.

4. Discussion

4.1. Co-creation

The set of 38 attributes, 10 principles, and three domains define co-creation as.

- A **collective intelligence process** that needs to be structured with systematic planning for *flexibility* to allow the collective intelligence. This needs to involve a *diversity* of co-creators. Diversity – the fuel of collective intelligence - is essential, not only because it is morally and ethically appropriate but also from a representativeness and inclusiveness perspective. More diverse teams of co-creators tend to be able to tackle more complex problems and find more creative solutions,[31] with a higher productivity and quality of the co-creation process.[32,33]
- This collective intelligence process must be underpinned by **scientific and methodological rigour**[34] aimed at answering a *clear and well-framed problem*[35] using the best available sources of *plural evidence*[36,37] and continuous *critical reflection*[36,37] and *evaluation*.[38,39]
- This is only possible if adequate **governance** is put in place to facilitate *an inclusive, open and transparent* process that acknowledges the contributions of the co-creators, prevents tokenism and ensures fundamental *ethical*[40]*and legal* requirements.

Consequently, to be trustworthy, co-creation should follow a structured though flexible protocol. The protocol should detail the governance, methodological rigour (including evaluation), and structure of the collective intelligence process, making sure that principles and attributes are maintained. Whilst we defined the attributes and principles of co-creation within the field of Public Health, we used literature on co-creation from all fields[8] and agnostic scientific methodology principles. Therefore, the attributes and principles apply to any fields and disciplines.

4.2. Attributes, principles, domains

Previous research tried to tackle confusion and vagueness around co-creation by providing definitions without a marked improvement in conceptual or operational clarity.[6] For example, many co-creation articles rightly state that co-creation should not be tokenistic but without

putting in place tangible measures to avoid tokenism.[8] Therefore, Health CASCADE opted to provide a concise, essential, and streamlined set of attributes, principles and domains that are fundamental to co-creation.

The advantage of this approach is that it delineates attributes that can be operationalised and upheld in practice. First, it promotes that all participants in the co-creation process operate with a shared understanding of its core attributes. The attributes constitute a fuller and pragmatic characterisation of co-creation, which fosters alignment, protects the integrity of the process and reduces miscommunication or misunderstandings. Second, the attributes are actionable and thereby guide the planning and delivery of co-creation,[41,42] both simplifying decision-making and also preventing tokenism and unwanted outcomes. Finally, the attributes can also inform process evaluations,[38,39] key performance indicators and metrics for success, allowing for better tracking of progress and outcomes during and after the co-creation process.

The results are inline with parallel conceptual work done by Pérez Jolles et al. (2022)[43] in implementation science who derived 5 broad principles for co-creation (Equity, Reflexivity, Reciprocity and Mutuality, Transformative and Personalized, and Facilitated relationships). These align directly with our governance domain (Equity, Transparency) and collective intelligence processes (Diversity, Structured relationships). This convergence of independent approaches strengthens the validity of these core co-creation domains. The attributes, derived in our study, additionally provide the scientific and methodological rigour domain with principles for clear problem definition, plural evidence, critical reflection, and rigorous evaluation that are essential for establishing co-creation as a trustworthy evidence-based methodology.

There were three notable differentiae (Empowerment/disempowerment, Unbiased/biased, and Political/apolitical - and the converse apolitical/political) which were omitted from our final principles, although they were hotly debated and were the hardest to reach a decision on. Although excluded from our principles, they are relatively common notions in the wider co-creation, co-production and co-design literature,[8] and perhaps require some explanation. Empowerment, for example, is commonly found in definitions of co-creation, co-production, and co-design being described as an "Empowering" approach. Yet, empowerment is a controversial topic with the potential for paradoxical effects.[44,45] As a methodology, co-creation should confer to the co-creators a greater sense of shared control over tackling a problem at hand and aim to empower. However, empowerment is more a desirable outcome of the process than an intrinsic attribute.

Another controversial differentiae was (Unbiased/biased). While science thrives to be unbiased, Co-creation brings multiple perspectives and knowledge together and is best underpinned theoretically by critical realism.[36] Therefore, co-creation thrives to compose solutions, from a diversity of biased perspectives and knowledges[46] purposely brought together. Consequently, it is probably wiser and more logical to openly report and map the biases at play through the attribute and principle of transparency rather than uphold an unachievable and unreachable idealist characteristic.

Similarly, it seems difficult to fathom how the scientific virtue of political neutrality, which is rarely realised in practice, could be a fundamental characteristic of co-creation. Therefore, the Apolitical/political differentiae was excluded. Indeed, co-creation is action-based and mission-orientated, which are both by nature political. Co-creation should not tend toward trying to achieve a consensus but instead should not be afraid of open controversy and debate that results in a diversity of solutions as there is rarely a one-size-fits-all solution in Public Health or Planatary Health.

4.3. Strengths and limitations

The main strength of this study is the innovative as well as rigorous, structured, and transparent collective intelligence process involving a wide diversity of actors. We are not aware of any previous study that has attempted to define the main attributes and guiding principles of co-creation. The process used a combination of methods, systematic feedback loops and member checking for cross-referencing and validation. The network devoted ample time to this task and purposely left plenty of time for reflection, stress testing and field testing between actions and phases. The network also allowed sufficient time to debate, hear different perspectives and allow controversies to be voiced and integrated. Finally, our "consent" approach is also a strength, as the purpose was to delineate and characterise co-creation but not to reach a "consensus" amongst ourselves and ensure that members of our network fully agree on every point. A consent approach reduces "Groupthink".[26]

However, the study has some limitations. First, despite our best efforts to base our process on extensive literature[8] and engage a wide range of actors, our proposed attributes and principles might have missed some information. Second, while contributors in the process continually field and stress-tested the attribute in practice in several fields of application, the full set of attributes and principles has only been used in practice for a year.

4.4. Practice and research implications

The 38 attributes and 10 principles provide actionable guidance for both practitioners and researchers conducting co-creation across all fields and contexts. In practice, these principles should be implemented and upheld throughout the entire co-creation lifecycle—from initial planning and stakeholder engagement, through active collaboration and knowledge generation, to final evaluation and dissemination. By adhering to these evidence-based principles, practitioners work toward a stronger assurance that their co-creation processes are trustworthy, transparent, and methodologically sound, whether addressing public health challenges in rural communities, developing implementation strategies in healthcare systems, or tackling urban planning issues in diverse cultural contexts. For researchers, these principles establish co-creation as a rigorous scientific methodology combining Mode 1 and Mode 2 that can be systematically applied, evaluated, and replicated across disciplines while maintaining scientific integrity. Rather than viewing co-creation as field-specific or context-dependent, our universal principles enable researchers to conduct methodologically robust co-creation that could generates credible evidence regardless of geographical location, cultural setting, or disciplinary domain. Future research should focus on developing standardised protocols and evaluation metrics based on these principles, creating training materials for practitioners, and conducting empirical studies to measure the impact of principle-adherent co-creation on both process quality and outcome effectiveness across diverse applications.

4.5. Conclusion

This study defines and operationalises co-creation as a robust scientific methodology and has led to the identification of 38 fundamental attributes, representing 10 guiding principles, within three domains. These elements collectively clarify co-creation as a transparent, structured, flexible process that engages a diverse group of co-creators. It should focus on a clear problem, integrate plural evidence, be reflective and include rigorous evaluation. Finally, co-creation should prioritise open access to information, recognise the contribution of co-creators and adhere to sound ethical standards. Practitioners and researchers should use and uphold these attributes, principles and domains during the planning, delivery and evaluation of co-creation for high-quality application of co-creation in public health and beyond, ultimately delivering meaningful, sustainable, and equitable outcomes.

Author statements

Ethical approval

Not required. This was a theoretical and literature review study with no personal data taken or experimental design on human or animals.

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Competing interests

The authors declared no conflict of interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT, Claude and Grammarly in order to help with dyslexia, and improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.puhe.2025.105922.

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